

DETERMINANTS & FUNDAMENTAL PROPERTIES

1. $\det(A) = \det(A^T) \implies |A^T| = |A|$
2. If A is a square matrix of order $n \times n$ and $c \in \mathbb{R}$ is any scalar, then:
 $\det(cA) = c^n \cdot \det(A)$
3. $\det(AB) = \det(A) \cdot \det(B)$

IMPORTANT NOTE ON MATRIX MULTIPLICATION

In general, matrix multiplication is non-commutative: $AB \neq BA$.

However, their determinants satisfy absolute equality:

$$|AB| = |BA|$$

| Proof Hook: $|AB| = |A||B| = |B||A| = |BA|$ (Since scalar values commute freely).

4. $\det(A + B) \neq \det(A) + \det(B)$ in general.

ILLUSTRATIVE EXAMPLE

Let A be a 3×3 matrix such that $\det(A) = 5$. Find $\det(-4A)$.

$$\det(-4A) = (-4)^3 \cdot \det(A)$$

$$\implies (-64) \times 5 = -320$$

ADJOINT AND INVERSE MATRIX IDENTITIES

Let A be a square matrix of order $n \times n$. The following identities hold firmly:

- | | |
|--|---|
| 5. $\text{adj}(A^T) = (\text{adj } A)^T$ | 9. $ \text{adj } A = A ^{n-1}$ |
| 6. $\text{adj}(cA) = c^{n-1} \cdot \text{adj } A$ | 10. $ \text{adj}(\text{adj } A) = A ^{(n-1)^2}$ |
| 7. $\text{adj}(\text{adj } A) = A ^{n-2} \cdot A$ | 11. $ A^{-1} = \frac{1}{ A }$ |
| 8. $A \cdot (\text{adj } A) = (\text{adj } A) \cdot A = A \cdot I$ | 12. $A^{-1} = \frac{1}{ A } \cdot \text{adj } A$ |

OPERATIONAL SHORTCUTS

$$\cdot (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

$$\cdot (\mathbf{A}^{-1})^{-1} = \mathbf{A}$$

$$\cdot \text{adj}(\mathbf{AB}) = (\text{adj } \mathbf{B})(\text{adj } \mathbf{A})$$

$$\cdot (\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$$

ORTHOGONAL MATRICES

A square matrix $\mathbf{A}$ of order $n \times n$ is defined as an **orthogonal matrix** if and only if:

$$\mathbf{A} \cdot \mathbf{A}^T = \mathbf{A}^T \cdot \mathbf{A} = \mathbf{I}_n$$

Key Algebraic Properties:

- Taking determinants on both sides:
 $|\mathbf{A} \cdot \mathbf{A}^T| = |\mathbf{I}| \implies |\mathbf{A}| \cdot |\mathbf{A}^T| = 1 \implies |\mathbf{A}|^2 = 1 \implies |\mathbf{A}| = \pm 1.$
- If $|\mathbf{A}| = 1$, the matrix is specifically classified as a **Proper Orthogonal Matrix**.

THE 2D STANDARD ROTATION MATRIX CASE

Let $\mathbf{A} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$. This matrix is proper orthogonal ($|\mathbf{A}| = 1$).

General Matrix Power Property:

$$\mathbf{A}^n = \begin{bmatrix} \cos(n\alpha) & -\sin(n\alpha) \\ \sin(n\alpha) & \cos(n\alpha) \end{bmatrix}$$

Vector Interpretation (Column Vectors): Let columns be vectors $\vec{c}_1 = [\cos \alpha, \sin \alpha]^T$ and $\vec{c}_2 = [-\sin \alpha, \cos \alpha]^T$.

1. **Orthogonality Check:** $\vec{c}_1 \cdot \vec{c}_2 = \cos \alpha(-\sin \alpha) + \sin \alpha \cos \alpha = 0$

2. **Normality Check:** $|\vec{c}_1| = \sqrt{\cos^2 \alpha + \sin^2 \alpha} = 1$, and $|\vec{c}_2| = \sqrt{(-\sin \alpha)^2 + \cos^2 \alpha} = 1$

IDEMPOTENT MATRICES

A square matrix A is classified as **idempotent** if and only if squaring the matrix yields itself:

$$A^2 = A$$

THEOREM: SIMULTANEOUS IDEMPOTENCY

If two matrices A and B satisfy the relations $AB = B$ and $BA = A$, then both A and B are independently idempotent:

$$A^2 = A \quad \text{and} \quad B^2 = B$$

Furthermore, their sums satisfy the relation: $A^2 + B^2 = A + B$

THEOREM: IDENTITY COMPLEMENT MATRIX

If A is an idempotent matrix ($A^2 = A$), then its identity complement matrix ($I - A$) is also idempotent.

Proof Verification:

$$\begin{aligned}(I - A)^2 &= (I - A)(I - A) = I^2 - I \cdot A - A \cdot I + A^2 \\ &= I - 2A + A^2\end{aligned}$$

$$\text{Since } A^2 = A \implies I - 2A + A = I - A$$

STANDARD CYCLIC DETERMINANT EQUATIONS

These structural cyclic patterns frequently surface in evaluations and should be evaluated directly:

I. TYPE 1: STANDARD LINEAR-SQUARE VANDERMONDE FORM

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a - b)(b - c)(c - a)$$

II. TYPE 2: SQUARE-CUBE FORM (LINEAR ROW EXCLUDED)

$$\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(ab + bc + ca)$$

III. TYPE 3: LINEAR-CUBE FORM (SQUARE ROW EXCLUDED)

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)$$

IV. TYPE 4: PURE CYCLIC SYMMETRIC CUBIC STRUCTURE

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc)$$

CHARACTERISTIC EQUATIONS VIA CAYLEY-HAMILTON THEOREM

Every square matrix A structurally satisfies its own unique characteristic equation, yielding critical shortcuts:

SHORTCUT FORMULA FOR 2×2 MATRICES

$$A^2 - (\text{tr } A)A + (\det A)I = \mathbf{O}$$

*Note: $\text{tr } A$ (Trace of A) = Sum of principal diagonal elements = $a_{11} + a_{22}$.

SHORTCUT FORMULA FOR 3×3 MATRICES

$$A^3 - (\text{tr } A)A^2 + (A_{11} + A_{22} + A_{33})A - (\det A)I = \mathbf{O}$$

Where A_{11}, A_{22}, A_{33} represent the principle diagonal cofactors of matrix A .

Alternatively defined as: $(A_{11} + A_{22} + A_{33}) = \text{trace}(\text{adj } A)$.